All-Optical Cross-Absorption-Modulation Based Gb/s Switching With Silicon Quantum Dots

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***Abstract*—In silicon quantum dots (Si-QDs), the sub-bandgap cross-absorption-modulation (XAM) has been preliminarily observed and confirmed as a new kind of wavelength conversion process to enable ultrafast optical switching. By using the Si-QD doped Si-rich SiC_x micro-ring resonator waveguide with a quality factor of 1.7×10^4 under an optimized gap spacing of 700 nm away from the bus waveguide, the XAM effect induces a wavelength-converted picosecond all-optical switching between pump and probe signals, which is attributed to a specific free-carrier absorption at probe wavelength caused by the two-photon-absorption induced carriers. The non-degenerate pump-probe analysis also shows a weak Kerr nonlinearity related ultrafast switching response from the changing envelope of the XAM probe pulse at red-shifted wavelength. These observations declare that the nano-scale Si-QDs can provide sufficiently large XAM effect to enable the ultrafast all-optical switching capability for pulsed optical logic applications. To confirm, the Si-QDs doped Si-rich SiC_x waveguide modulator-based 1.2 Gb/s all-optical format inversion of a PRZ-OOK data-stream is demonstrated for the first time.**

***Index Terms*—Cross-absorption-modulation, silicon carbide waveguide, Si quantum dots.**

I. Introduction

DEVELOPING ultrafast speed data processing for silicon photonics is one of the most compelling integrated circuits need in next decades [1]. To brighten the prospect demands, a lot of effort has focused on emerging new class of ultrahigh speed Si-based optical switches and modulators. Typically, the long minority carrier lifetime existed in bulk Si (μs-ms) inevitably limits its modulation bandwidth in few MHz. In addition to the carrier diffusion process, the carrier lifetime shortening usually relies on either enlarging the drift velocity or shortening the lifetime of reacting carriers in Si-based photonic device. New concepts and approaches were successively proposed to enhance the speed and bandwidth in Si photonic devices [2]–[6]. With the use of dense carrier trapping center, Wright *et al*. reported the carrier lifetime reduction of Si waveguide to 0.57 ns via silicon ion implantation in 2008 [5]. Alternatively, Turner-Foster *et al.* further shortened the drift time of free carrier in Si nano-scale waveguide (to <12.2 ps) by introducing electric field through an integrated p-i-n diode to effectively sweep out carriers [6]. In previous works, the Si-based optical modulators fabricated by crystalline and polycrystalline Si were generally demonstrated by using effects of free carrier dispersion and free carrier absorption (FCA) effect, which enable several kinds of functionalities such as all-optical switching [7], thermo-optic switching [8], electro-optic modulation [9]–[11], phase shifting [12] and optical logics [13].

Later on, a revolutional design has emerged by embedding Si quantum dots (Si-QDs) into Si-based matrices to shorten the carrier lifetime and to enhance the absorption cross section due to its quantum confinement effect (QCE). The QCE induced by Si-QD broadens the wave function of e-h pairs to facilitate for the formation of the quasi-direct bandgap and to stimulate the carrier recombination without assistance of phonon [14], [15]. Therefore, the Si-QD based optoelectronics have been developed in recent years [16]–[21]. With this benefit, Kekatpure *et al.* investigated that the FCA cross-section in Si-QD is one order of magnitude larger than that of bulk Si under strong QCE. Therefore, the all-optical FCA modulator fabricated with embedded Si-QDs is expected to possess higher modulation depth than that with bulk Si [22]. Marchena *et al.* demonstrated the FCA induced loss in the Si-QD based micro-disk at resonant frequency by optical pumping [23]. With the FCA effect in Si-QD doped SiO_x waveguide modulator, the all-optical switching was performed by employing a pulsed pump laser to control the transient free-carrier density so as to rapidly modulate the probe signal passing through the Si-QD [24]–[26]. In principle, the FCA cross-section of the Si-QD can be affected by the QCE, such that the free carrier lifetime as well as the FCA modulation bandwidth can be manipulated by controlling the size of the Si-QD. The latter works have declared that the bandgap property as well as carrier lifetime of Si-QD can be significantly modified by shrinking the Si-QD size [14], [27]. For example, the carrier lifetime can be significantly shortened to ∼ns by introducing QCE effect and defect state in the SiO_2 or SiN_x matrix [28]–[30], which benefits from the improvement on the modulation bandwidth to 10–100 s MHz. Nevertheless, the practical application is still limited due to the isolated electrical property of SiO_x host matrix which restricts the fast variation on carrier density. To sweep out excess carriers for achieving high-speed modulation, the SiO_x host matrix should be replaced by the semi-conducting material such that the carrier diffusing

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capability can be enhanced. More recently, the semiconducting SiC_x matrix has been employed a nice candidate due to its structural stability under high-power operation [31], short bulk carrier lifetime [32], low absorption coefficient at 1550 nm and large refractive index. These advantages make the silicon rich (Si-rich) SiC_x, an expectable matrix to host the tiny Si-QDs for enlarging the modulation bandwidth.

In this work, the Si-QD doped Si-rich SiC_x (SiC_x:Si-QD) waveguide based picosecond all-optical cross-absorption-modulation (XAM) switching is demonstrated. The quality factor and modulation depth of the all-optical XAM switch can be improved by optimizing the gap spacing between the bus and ring waveguides. The carrier lifetime and dispersion factor of SiC_x:Si-QD are analyzed by the modulated probe signal. To distinguish the nonlinear Kerr effect and the TPA induced XAM effect in the SiC_x:Si-QD waveguide, the switching response is numerically simulated for decomposing the individual contribution from both mechanisms. The TPA induced free carriers result in the FCA in the SiC_x:Si-QD modulator to dominate the all-optical data conversion for delivering picosecond switching with pulsed-return-to-zero on-off keying (PRZ-OOK) data format.

II. Experimental Setup

A. *Structural and Compositional Characteristics of Si-Rich SiC_x Film*

The Si-rich SiC_x film was deposited on SiO_2 covered Si substrate by using the plasma-enhanced chemical vapor deposition (PECVD) system with reactive gaseous recipe of argon diluted silane (8% SiH_4 + 92% Ar) and methane (CH_4) [33]. During synthesis, the substrate temperature and chamber pressure were set as 550 °C and 0.3 torr, respectively. The RF plasma power and the deposited time were maintained as 120 W and 10 min, respectively. The X-ray photoelectron spectroscopy (XPS) with an Mg K_α-line radiation source was used to characterize the atomic composition ratio of the Si-rich SiC_x film. The Raman scattering spectroscopy was utilized to analyze bonding and the crystallization of the Si-rich SiC_x. The coupling factor at the bus-ring output port also reveals an inverse trend with the Q-factor, indicating that the optimized gap spacing must be set as 700 nm to compensate the enlarged Q-factor and the reduced coupling efficiency.

The surface images of SiC_x:Si-QD micro-ring resonator waveguide are shown in Fig. 1(a). In this bar, there are totally 40 sets of SiC_x:Si-QD resonator waveguides aligned in parallel with each waveguide length of 3 mm. Fig. 1(b) shows the XPS spectrum of the Si-rich SiC_x film. The Si_{2p} signal is observed at 99.8 eV and the C_{1s} signal is observed at 281.9 eV. The atomic ratios of Si and C are 59.4% and 32%, respectively. In comparison with the stoichiometric SiC with a C/Si ratio of 1, the C/Si ratio for the synthesized Si-rich SiC_x of 0.54 gives rise to a molar fraction of $x = 0.54$ and an excess Si concentration of 27.4% in Si-rich SiC_x film. In general, the excessive Si atoms are self-assembled to form a-Si clusters which are randomly distributed in the as-deposited SiC_x films. The decomposition of SiH_4 molecules is occurred at lower temperature due to its lower dissociation energy (75.6 kcal/mol) [34] than that of CH_4 molecule.

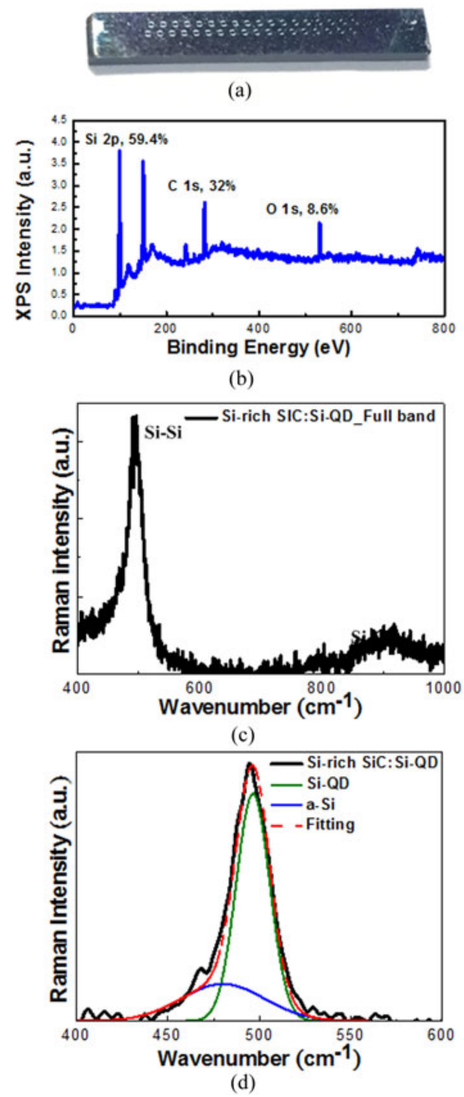

Fig. 1. (a) Photographs on surface images of the SiC_x:Si-QD micro-ring resonator waveguide. (b) The XPS spectrum of Si-rich SiC_x film. (c) and (d) The Raman spectrum of Si-rich SiC_x, and the decomposed Si–Si related Raman scattering peaks.

This facilitates obtaining the Si-rich condition in SiC_x films with low-temperature deposition process in PECVD system.

To characterize the bonding structure and crystallinity, the Raman scattering spectroscopy of Si-rich SiC_x film was performed to analyze the crystallization of SiC_x matrix and Si-QDs self-assembled from excessive Si atoms, as shown in Fig. 1(c). The Raman signal of Si-C longitudinal optic (LO) phonon mode located at 973 cm^{-1} is unobservable; however, another red-shifted observed LO mode at 917 cm^{-1} with a broadened linewidth presents due to the existence of SiC nano-clusters [35], [36]. In comparison with the crystalline Si related scattering Raman signal at 520 cm^{-1}, the observed peak of Si-rich SiC_x is shifted to 495 cm^{-1} with a broadened linewidth of 24.7 cm^{-1} due to the QCE of Si-QD [37], [38]. In detail, the deconvolution of Si related peaks are performed by fitting with multiple Gaussian functions to further analyze the bonding composition, as shown in Fig. 1(d). Two components are observed from the

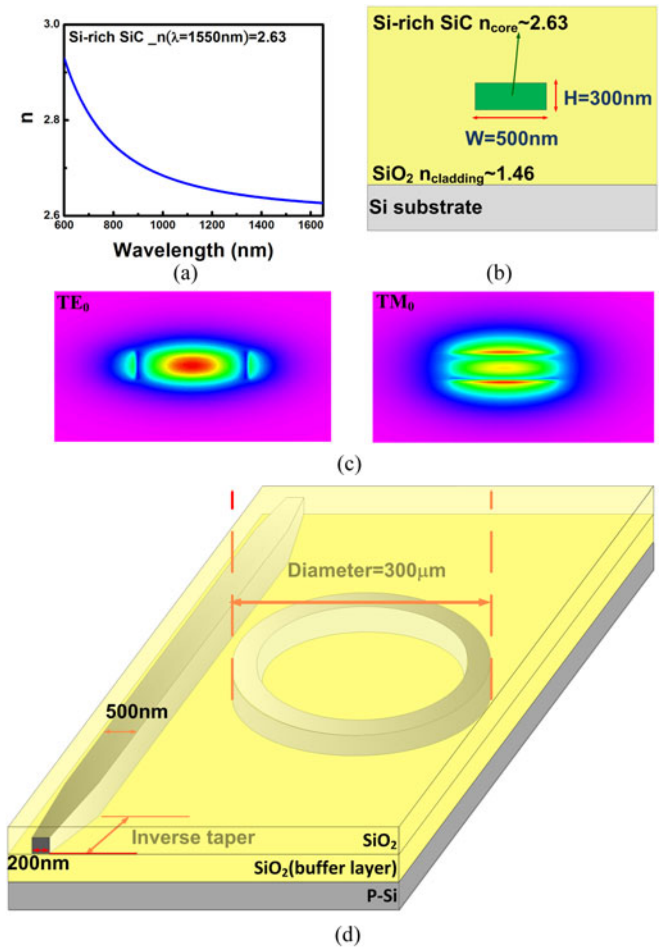

Fig. 2. (a) The refractive index spectrum of Si-rich SiC_x films deposited on Si substrate. (b) The cross-section view of the SiCx:Si-QD based channel waveguide. (c) The simulation of transverse modes for TE_0 and TM_0 modes. (d) The scheme of SiC_x:Si-QD micro-ring channel waveguide.

decomposed Raman spectrum, including the dominant Si-QD related peak located at 496 cm^{-1} with corresponding FWHM of 21.6 cm^{-1}, and the amorphous Si related peak at 480 cm^{-1} with corresponding FWHM of 53 cm^{-1}. As confirmed by the Raman scattering analysis of Si-rich SiC_x, both the amorphous Si clusters and Si-QDs can self-assemble in the Si-rich SiC_x films.

B. Design and Fabrication of the SiC_x:Si-QD Micro-Ring Resonator

To determine the geometric structure of the Si-rich SiC_x based micro-ring resonator, the refractive index of Si-rich SiC_x film was firstly measured by using the ellipsometer, as shown in Fig. 2(a). The refractive index of Si-rich SiC_x at wavelength of 1550 nm is ~2.63. To ensure the single mode operation in the Si-rich SiC_x based channel waveguide, the R-soft software with a beam propagation method is used to determine the geometric structure [39]. Fig. 2(b) provides the cross-section view of the Si-rich SiC_x based channel waveguide, and the refractive indices of core and cladding are set as 2.63 and 1.46, respectively. Afterwards, the width and height of the Si-rich SiC_x core layer are set as 500 nm and 300 nm, respectively. The simulated fundamental modes at wavelength of 1550 nm in SiC_x channel waveguide is shown in Fig. 2(c).

Fig. 2(d) demonstrates the configuration of the SiC_x:Si-QD based all-optical XAM switch. In the initial stage, the Si-rich SiC_x film with thickness of 300 nm on the SiO_2/Si substrate was fabricated by using PECVD system. Subsequently, the patterns of ring and bus waveguide were determined by the e-beam lithography on the positive photoresist, where the width of the waveguide was set as 500 nm. The gap spacing between the bus and micro-ring waveguide was determined as 700 nm, and the diameter of the micro-ring waveguide was designed as 300 μm. The inverse taper structure with length of 200 μm and enlarged width from 200 to 500 nm was introduced to enhance the power coupling into the waveguide [40]. After that, a Cr film with thickness of 80 nm was evaporated on the patterned wafer to serve as the hard mask, where the unpatterned region of SiC_x film was etched by using the reactive ion etching (RIE) with gaseous CF_4 to form the SiC_x:Si-QD based all-optical XAM switch. Afterwards, the Cr hard mask was removed by the etchant, and the 2-μm thickness of SiO_2 was deposited on the SiC_x based all-optical XAM switch to serve as the upper-cladding layer. Eventually, the waveguide was cleaved and polished to minimize the coupling loss at both end-facets.

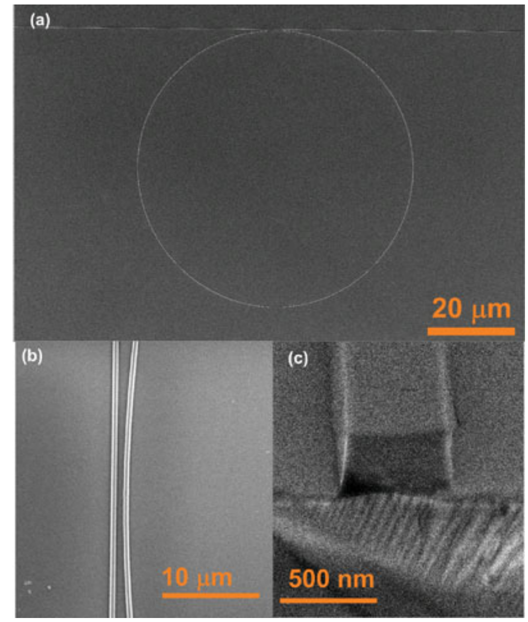

Fig. 3. The SEM images of the SiC_x micro-ring resonator and, including (a) the whole structure (b) the gap between the bus waveguide and micro-ring, and (c) the bus waveguide.

The SEM images of the SiC_x micro-ring waveguide are as shown in Fig. 3. The SiC_x micro-ring waveguide resonator with its radius of 150 μm is shown in Fig. 3(a). Fig. 3(b) exhibits the gap spacing between bus waveguide and micro-ring resonator of 700 nm. Fig. 3(c) displays the width and height of the SiC_x channel waveguide of 500 and 300 nm, respectively, to confirm the design. Fig. 4 illustrates the experimental setup of a pump-probe system for measuring the all-optical XAM switching of the SiC_x:Si-QD waveguide. To generate an

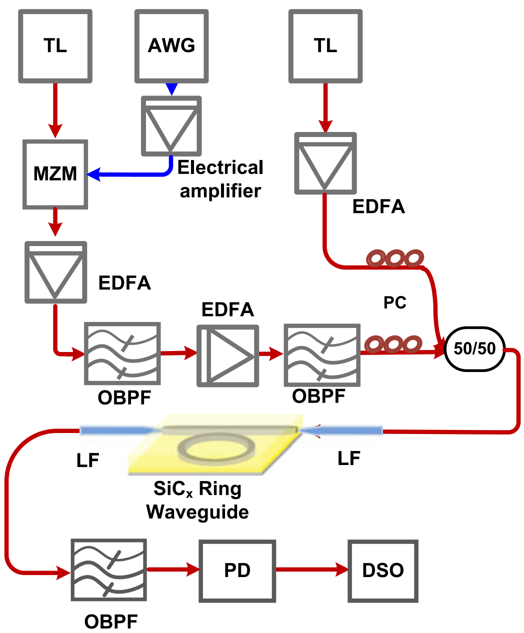

Fig. 4. A pump-probe system for measuring the all-optical cam switching of SiC_x:Si-QD based all-optical XAM switch resonator.

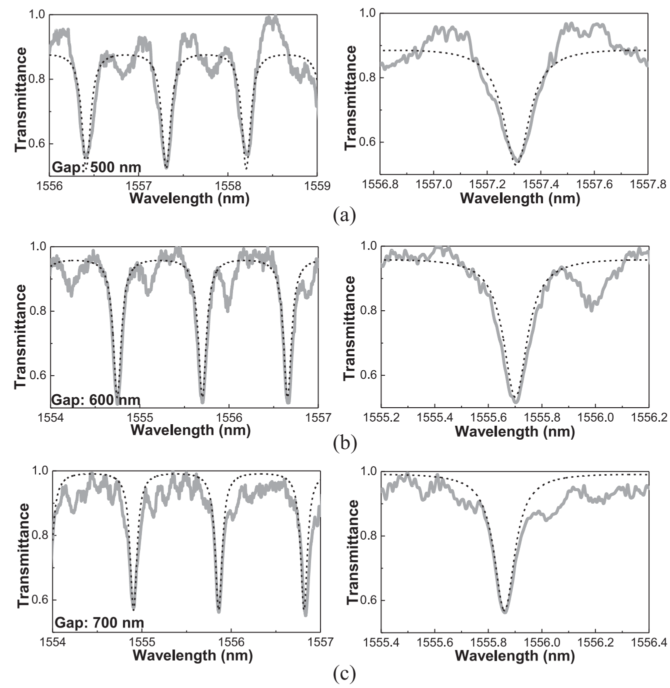

Fig. 5. The experimental and simulated transmission spectra of SiC_x:Si-QD bus and based all-optical XAM switch with different gap spacings between bus and based all-optical XAM switch. (a) $D_{\mathrm{gap}} = 500$ nm. (b) $D_{\mathrm{gap}} = 600$ nm. (c) $D_{\mathrm{gap}} = 700$ nm.

intensive pump pulse, a continuous wave (CW) tunable laser was externally modulated by an arbitrary waveform generator (AWG, Tektronix, 7122B) via a Mach-Zehnder modulator (MZM, JDSU, 10024180). In order to enhance the optical peak power of the modulated pump pulse, the signal was amplified by using a first-stage erbium doped fiber amplifier (EDFA, JDSU, OAB1552+20FA6). In addition, to eliminate the amplified spontaneous emission (ASE) noise induced by the EDFA, the amplified optical pulse was filtered by an optical bandpass filter (OBPF, SANTEC, OTF-910). In the next stage, the pump pulse was further strengthened by a second-stage EDFA (SDO, EFAH1B111NC02). The laser pulsewidth was 83 ps with a repetition rate of 12 MHz. The peak power of the optical pump pulse was enlarged to $P_{\mathrm{peak}} = 6.7$ W.

On the other hand, a probe beam from a CW tunable laser was amplified by an EDFA. A 3 dB coupler was utilized to combine the pump and probe signals for injecting them into the SiC_x based all-optical XAM switch via a lensed fiber (LF). A polarization controller was used to detune the polarization of the optical field. The wavelengths of the pump and probe signals were adjusted to the different resonance peaks of the micro-ring resonator. In order to distinguish the pump and probe wavelengths at the receiving port, an OBPF was used to eliminate the pump pulse so as to send the modulated probe signal into the photodetector. The electrical signal was further amplified to be detected by a digital sampling oscilloscope.

III. Results and Discussions

A. Transmission Spectra of SiC_x:Si-QD Micro-Ring Resonator With Different Gap Spacings

The normalized transmission spectrum of SiC_x:Si-QD micro-ring resonator can be numerically simulated by using the analytically derived equation as below [41],

$$\mathrm{T}(\lambda) = \left|\frac{\mathrm{A}}{\mathrm{A}_0}\right|^2 = 1 - \frac{(1-x^2)(1-y^2)}{(1-\mathrm{xy})^2 + 4\mathrm{xy}\sin^2(\pi \mathrm{n}_{\mathrm{eff}}\mathrm{L}/\lambda)}, \quad (1)$$

where $x = \exp(-\alpha_{\mathrm{ring}} L/2)$ and $y = \sin(\kappa l)$, A_0 denotes the input optical amplitude of the waveguide, A the output amplitude of the waveguide, α_{ring} denotes the propagation loss coefficient inside the micro-ring resonator and L is the cavity length, κ is the coupling coefficient, l is the effective coupling length between bus and based all-optical XAM switch, n_{eff} is the effective group refractive index of SiC_x:Si-QD based channel waveguide and λ is the central wavelength. After simulation, the fitted transmission spectrum of SiC_x:Si-QD based all-optical XAM switch by using Eq. (1) is shown in Fig. 5.

In addition, the transmission spectra ranged between 1563 and 1566 nm for SiC_x:Si-QD bus and based all-optical XAM switch with different gap spacings are shown in Fig. 6. The simulated group index is 2.6918, and the propagation loss in the micro-ring resonator is evaluated as 6 ± 0.5 cm^{-1}. By increasing the gap spacing (D_{gap}) between the bus and ring waveguide from 500 to 700 nm, the full-width at half maximum (FWHM) of the resonance dip on the SiC_x:Si-QD micro-ring resonator decreases from 116.4 to 90.2 pm, providing a quality (Q) factor of the micro-ring resonator enlarged from 1.34×10^4 to 1.72×10^4. The SiC_x:Si-QD bus and based all-optical XAM switch with a gap spacing of 700 nm exhibits the largest Q factor among all conditions. The fitting parameters of the SiC_x:Si-QD micro-ring resonator with different gap spacing are listed in Table I.

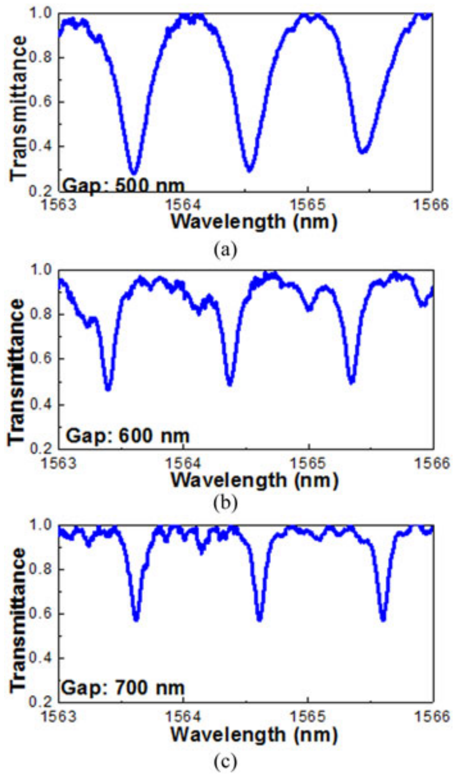

Fig. 6. The experimental spectra ranged between 1563 nm and 1566 nm for SiC_x:Si-QD bus and based all-optical XAM switch with different gap spacings between bus and based all-optical XAM switch. (a) $D_{\text{gap}} = 500$ nm. (b) $D_{\text{gap}} = 600$ nm. (c) $D_{\text{gap}} = 700$ nm.

TABLE I
THE OPTICAL PROPERTIES AND SIMULATED PARAMETERS OF SI-RICH SiC_x MICRO-RING RESONATOR

Sample	SiC_x:Si-QD		
D_{gap}, Gap spacing (μm)	0.5	0.6	0.7
R, radius of micro-ring waveguide (μm)		150	
N, group index		2.6918	
y, coupling factor	0.12	0.086	0.071
α_{ring} (cm^{-1})	15.07	13.91	12.75
$\delta\lambda$, FWHM (nm)	0.1164	0.1045	0.0902
$\Delta\lambda$, FSR (nm)	0.9596	0.9562	0.9531
Quality factor, Q	13355	14881	17248

To verify the optimized design, the commercially available beam-propagation-method program (R-soft) was utilized to simulate the Q-factor of the micro-ring resonator in the SiC_x:Si-QD based all-optical switch. At the beginning, a Gaussian beam was established to obtain the fundamental mode after propagating through the SiC_x:Si-QD bus waveguide, as illustrated in Fig. 7(a). When the simulated fundamental mode was inserted into the bus waveguide, the coupling factor versus propagating distance z can be obtained in Fig. 7(b).

Subsequently, the coupling factor and related parameters obtained from experiments (listed in Table I) were employed to calculate the Q-factor of the SiC_x:Si-QD micro-ring resonator with different gap spacing. The simulated and experimental Q-factors of the SiC_x:Si-QD micro-ring resonator with different gap spacing are also compared each other. The simulation interprets that the SiC_x:Si-QD based all-optical XAM switch with gap spacing separated from the bus waveguide possesses high Q-factors, but the increasing trend of Q value is saturated when the gap spacing exceeds beyond 700 nm. In principle, the Q-factor is obtained as below

$$Q = \frac{\pi\sqrt{x\sqrt{(1-y^2)}}Nl}{(1-x\sqrt{(1-y^2)})\lambda_0}. \tag{2}$$

The coupling factor is reduced when increasing the gap spacing. When increasing the gap spacing, the light is hardly coupled into the ring resonator, and the light in ring resonator is also hardly coupled back to the bus waveguide. This phenomenon contributes to the decreased cavity loss and enhanced Q factor; however, the coupling factor would decay to 0 with infinitely large gap spacing such that the Q factor saturates accordingly. In addition, the extinction ratio of the SiC_x based micro-ring resonator waveguide increases from 2.26 to 2.52 dB with enlarging the gap spacing from 500 to 700 nm. In experiment, the sidewall of the bus waveguide and ring resonator is slightly roughened after the ion-beam etching fabrication, which induces higher loss, lower Q-factor and smaller magnification factor than expectation. Therefore, the simulated Q-factor (coupling factor) is not equivalent to the experimental one.

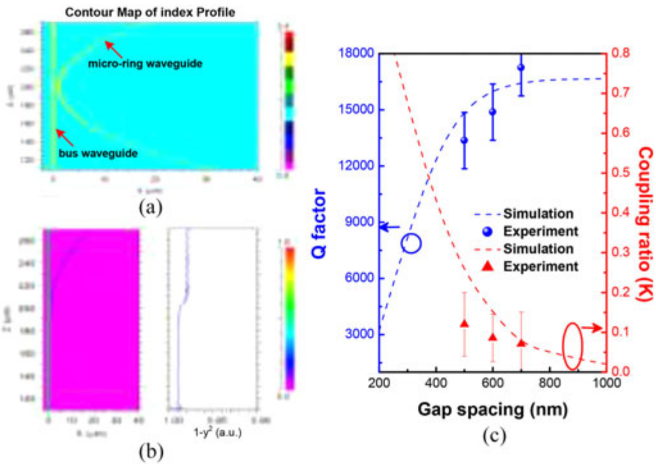

Fig. 7. (a) Refractive index profile of bus and based all-optical XAM switch. (b) The coupled optical intensity varied with the propagating length through the bus waveguide. (c) The simulated and experimental Q factor of SiC_x:Si-QD based all-optical XAM switch with different gap spacing.

B. XAM Switching and Carrier Lifetime Estimation in SiC_x:Si-QD Waveguide

Later on, the modulated traces of the probe signal with corresponding fitting curves at different wavelengths are shown in Fig. 8. Owing to the inhomogeneous size distribution of the Si-QDs [14], the free carrier lifetime dispersion by the Si-QDs will be induced to degrade the modulation response of the SiC_x:Si-QD based all-optical XAM switch. In principle, the transient free-carrier absorption induced probe power depletion with excited free carriers excited in the SiC_x:Si-QD based all-optical

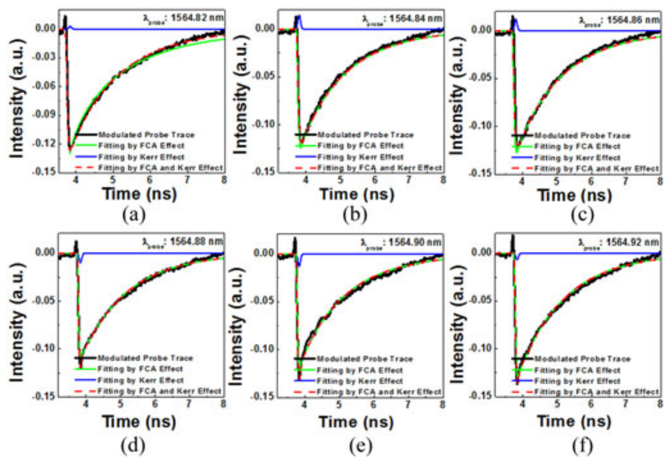

Fig. 8. The modulated probe beam in SiC_x:Si-QD micro-ring resonator with corresponding fitting curves of FCA and Kerr effects at different wavelengths. (a) $\lambda_{\text{probe}} : 1564.82$ nm, (b) $\lambda_{\text{probe}} : 1564.84$ nm, (c) $\lambda_{\text{probe}} : 1564.86$ nm, (d) $\lambda_{\text{probe}} : 1564.88$ nm (e) $\lambda_{\text{probe}} : 1564.90$ nm and (f) $\lambda_{\text{probe}} : 1564.92$ nm.

XAM switch can be fitted by using a stretched exponential decay function, as given by [14]:

$$P_{\text{FCA}}(t) = P_{\text{FCA}} \times \exp[-[t/\tau_c]^{\beta}] \tag{3}$$

where $P_{\text{FCA}}(t)$ denotes the probe beam profile in time-domain when cross-wavelength modulated by the FCA effect, P_{FCA} is the throughput probe power obtained after experiencing the XAM mechanism in the bus waveguide, τ_c is the carrier lifetime, and β is the dispersion factor of carrier lifetime distribution corresponding to the inhomogeneous broadening on Si-QD size distribution. The relaxation lifetime and dispersion factor are estimated as $\tau_c = 1.3$ ns and $\beta = 0.95$, respectively. Surprisingly, the experimentally obtained transient probe profile cannot be well fitted by only using Eq. (3).

In particular, the peak intensity of the transiently modulated probe waveform seems to exhibit an ultrafast response with varying intensity and sign at rising edge of the whole modulated profile. This problem can be numerically solved by incorporating a hyperbolic secant (sech) pulse into the simulation so as to perfectly fit the experimental result. Theoretically, the added pulse response in the fitting curve indicates that a weak nonlinear Kerr effect is possibly existed in the SiC_x:Si-QD micro-ring resonator under intense pumping. However, the contribution of nonlinear Kerr effect is weaker than that of FCA effect, which leads the FCA effect to dominate the cross-wavelength amplitude modulation in the SiC_x:Si-QD based all-optical XAM switch. Numerically, the added transient sech pulse profile $P_{\text{Kerr}}(t)$ induced by the nonlinear Kerr effect can be represented,

$$P_{\text{Kerr}}(t) = P_{Kerr} \exp\left[-\frac{t^2}{2 \times (\tau_p)^2}\right], \tag{4}$$

where P_{Kerr} is the throughput power of modulated probe response contributed by the nonlinear Kerr effect, τ_p is the FWHM of Kerr effect related response. The fitting parameters of modulated probe traces at different based all-optical switch in the SiC_x:Si-QD waveguide are listed in Table II. The positive or negative amplitude of the probe output which represents if the sign of the probe at different wavelengths is inversed by the pulsed pump data induced FCA or Kerr effect. The positive amplitude of the output probe means that the data is format preserved, whereas the negative amplitude of the output probe reveals the data format inversion.

TABLE II
THE SIMULATED PARAMETERS FROM TPA INDUCED FCA AND NONLINEAR KERR EFFECT AT DIFFERENT WAVELENGTH

Sample		SiC_x:Si-QD	
Wavelength (nm)	1564.82	1564.84	1564.86
P_{FCA}	−0.13	−0.125	−0.13
τ_c (ns)		1.3	
P_{Kerr}	0.003	0.015	0.012
τ_p (ps)		83	
Wavelength (nm)	1564.88	1564.90	1564.92
P_{FCA}	−0.11	−0.12	−0.13
τ_c (ns)		1.3	
P_{Kerr}	−0.01	−0.013	−0.07
τ_p (ps)		83	

During simulation, the modulated probe intensity gradually increases its intensity when red-shifted its wavelength from 1564.82 to 1564.84 nm. Conversely, the modulated probe intensity is reduced by further shifting its wavelength from 1564.84 to 1564.86 nm. As the probe wavelength approaches to 1564.88 nm, the sign of the Kerr effect related pulse response inverted modulation of the probe beam is observed. The largest intensity of the sign inverted response is occurred at wavelength of 1564.90 nm. Within such narrow detuning wavelength range, the nonlinear Kerr effect is existed to cause an intensity varied and sign reversed sech pulse response in the SiC_x:Si-QD based all-optical XAM switch. Nevertheless, the contribution of nonlinear Kerr effect is weaker than that of the TPA induced FCA effect so as to make the FCA effect a dominant factor in the SiC_x:Si-QD based all-optical XAM switch. As expected, the initial TPA and successive FCA induced XAM effect in the SiC_x:Si-QD based all-optical switch can be observed under high intensity pumping. When the total energy of two pump photons exceeds the bandgap energy of the Si-rich SiC_x, the TPA effect inside the SiC_x occurs to absorb the pump photons simultaneously, which is called the phonon-assisted degenerate TPA process [42].

Alternatively, the non-degenerate TPA process will also take place as long as the total energy of pump photon incorporated with another probe photon becomes larger than the bandgap energy of Si-rich SiC_x. The degenerate and non-degenerate TPA processes can provide free carriers from valence band to conduction band under intense pumping. The induced free carriers in the SiC_x:Si-QD would effectively absorb the probe photons so as to reduce the throughout intensity of probe beam. The Fig. 9 plots the output versus input power of the pumping signal at wavelength of 1556.7 nm. The output should be linearly proportional to the input without the TPA induced FCA effect. The pumping output saturates by enlarging the pumping intensity beyond 0.65 GW/cm^2 as two pumping photons was concurrently absorbed and transformed to an electron. This decreases

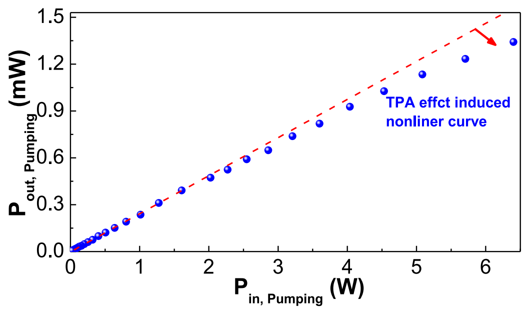

Fig. 9. The output versus input power of the pumping signal at wavelength of 1556.7 nm in the SiC_x:Si-QD based all-optical switch.

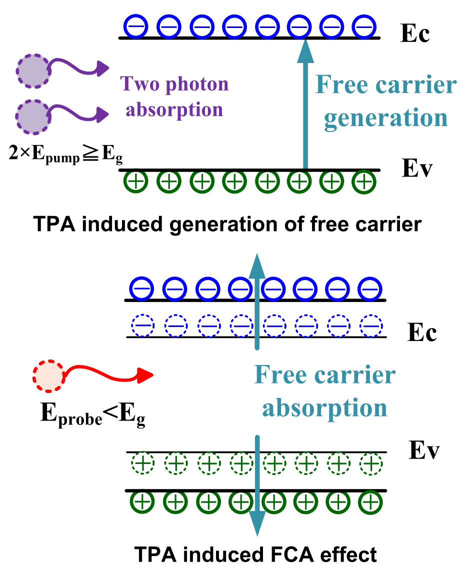

Fig. 10. Upper: The TPA effect induced free carrier in the SiC_x:Si-QD. Lower: TPA effect induced FCA absorption in the SiC_x:Si-QD.

the output pumping power as the nonlinearly TPA is gradually enhanced with increased pumping intensity.

C. Hybrid Strong XAM and Weak Kerr Switching Effects in SiC_x:Si-QD Based All-Optical Switching

In principle, the XAM effect can be described as the contribution from TPA induced FCA process in the SiC_x:Si-QD waveguide, as schematically illustrated in Fig. 10. Under sub-bandgap high-intensity pumping, the total energy of two photons successively absorbed by one carrier in valence band exceeds the forbidden bandgap energy of the Si-QD in Si-rich SiC_x. The TPA effect occurs inside the Si-QD to generated free-carriers, and the free-carriers continuously absorb the probing photons to cause FCA effect at another wavelength. This eventually leads to the XAM effect by transiently exciting free-carriers via TPA and re-absorbing the free-carrier with FCA.

In the meantime, the output of the probing signal will be inversely converted by the XAM effect. On the other hand, either the free carriers or the tremendous pump can induce nonlinear refractive index change, which could possibly introduce another switching response if the nonlinearity originated from afore-

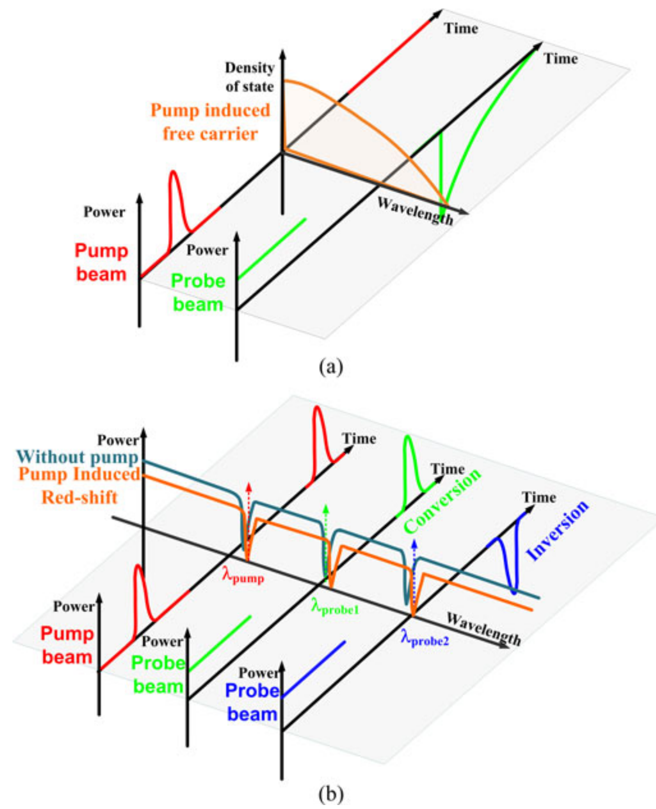

Fig. 11. The schematic diagrams of all-optical switching in the SiC_x:Si-QD based all-optical XAM switch induced by (a) FCA and (b) nonlinear Kerr effects.

mentioned effects is sufficiently large. Therefore, both kinds of optical switching responses need to be characterized concurrently, which helps to distinguish the XAM switching so as to realize the dominated mechanism inside the SiC_x:Si-QD based all-optical XAM switch. As a result, the probe beam can be inversely modulated by the pump pulse with the TPA induced FCA effect. This is so-called XAM effect and the schematic diagram of TPA-FCA induced XAM switching is shown in Fig. 11(a). In contrast, the Fig. 11(b) shows another switching mechanism based on the purely nonlinear Kerr effect occurred in the SiC_x:Si-QD micro-ring resonator under intense pumping condition. Originally, the wavelength of pump beam is set at the resonance dip of the transmission spectrum of the SiC_x:Si-QD micro-ring resonator to obtain the maximal power coupled into the micro-ring cavity for inducing the largest refractive index change.

Under intense pumping, the resonant micro-ring cavity transmission dips on the throughput of bus waveguide are red-shifted due to the increased refractive index in the SiC_x:Si-QD based all-optical XAM switch. Therefore, the transmittance of probe beam could also be changed due to the nonlinear Kerr effect induced transient red-shift on the transmission notch spectrum. To implement a direct all-optical modulation, the wavelength of probe beam is located at the original resonant dip of the SiC_x:Si-QD micro-ring, the injected pump pulse then red-shifts the transmission spectrum to increase the probe beam intensity. In contrast, when the wavelength slightly deviates from the

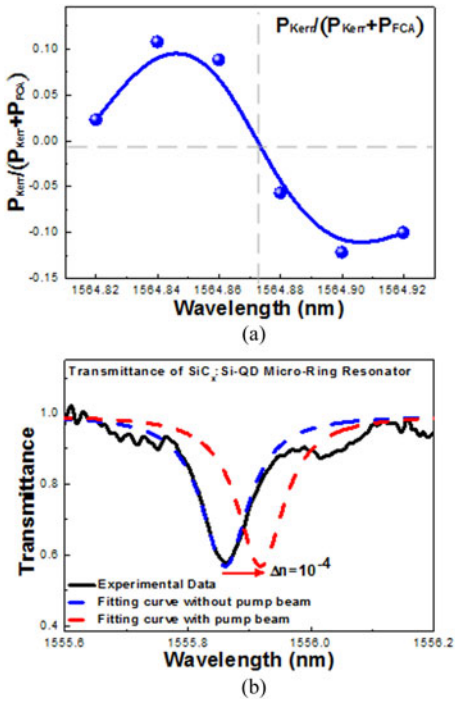

Fig. 12. (a) The power ratio of modulated probe beam contributed by nonlinear Kerr effect $(P_{\rm Kerr})$ and FCA effect $(P_{\rm FCA})$ at different wavelength. (b) The transmission spectra of SiC_x:Si-QD micro-ring resonator with and without pumping.

resonant dip, the injected pump pulse red-shifted transmission spectrum inversely decreases the probe beam intensity. Based on the working principle through modifying the refractive index of the SiC_x:Si-QD in the micro-ring, the probe beam can be directly all-optical modulated with either preserved or inverted data format by injecting high-power pump beam.

To quantitatively scale the effect of nonlinear Kerr switching on the probe output, the power ratio of nonlinearly Kerr switched power to the total power of the probe beam which denotes as $[P_{\rm Kerr}/(P_{\rm FCA}+P_{\rm Kerr})]$ at different wavelengths is shown in Fig. 12(a). As the Kerr effect takes place at very beginning (within 83 ps of the pump pulse duration), the Kerr induced spectral red-shift can be distinguished from the FCA induced absorption change and blue-shifted notch occurred at subsequent 8 ns or longer. Indeed, only the Kerr effect can be resolved at the rising part of the measured probe, while the effects of FCA caused spectral shift and absorption variation mix up each other. In particular, the absorption change gives rise to a negative peak in probe with its amplitude ratio enlarged up to 90%, which makes the spectral-shift induced intensity variation difficult to be separated from each other. At current stage, only the Kerr induced spectral shift and intensity variation can be distinguished from those induced by the FCA effect.

As a result, the spacing between wavelength at positive maximal power ratio of 0.1 (at 1564.84 nm) and that at negative maximal power ratio of -0.12 (at 1564.90 nm) is about $\sim$0.06 nm, indicating that the resonant transmission dip of the micro-ring resonator must be red-shifted by 0.06 nm when illuminating the transient pump pulse passing through upon the ring resonator. Theoretically, the nonlinear refractive index of SiC_x:Si-QD induced by the intense pulsed pump can be determined from analyzing the red-shifted wavelength of the resonant dip on the SiC_x:Si-QD micro-ring resonator. The effect of nonlinear refractive index on the transmittance of micro-ring resonator can be expressed by rewriting the $\mathrm{n_{eff}}$ in Eq. (1) as

$$n'_{\rm eff} = n_{\rm eff} + n_2 I_A(t) \tag{5}$$

where n_2 denotes the nonlinear refractive index of the Si-rich SiC_x, I_A the amplified peak intensity circulated in the micro-ring resonator. When the product of nonlinear refractive index and pump intensity is sufficiently large to induce a significant nonlinear Kerr switching for changing the refractive index of micro-ring resonator, the transmission spectrum of the SiC_x:Si-QD micro-ring resonator is red-shifted. Fig. 12(b) shows the simulated transmission spectra with and without pump. The red-shifted wavelength of $\Delta\lambda = 0.06$ nm is correlated with the group index change $\Delta n = n_2 I_A = 1\times10^{-4}$ as induced by the nonlinear Kerr effect. This gives rise to a nonlinear refractive index (n_2) of the Si-rich SiC_x as $n_2 = \Delta n/I_A$. Assuming that the peak power of the pump pulse amplified by two-stage EDFA is $\sim$6.7 W.

The insertion loss of -10 dB is defined as the total loss contributed by the 1×2 coupler, LF and the coupling loss per waveguide facet. That is, the residual peak power injected into the waveguide is approximately 0.67 W. On the other hand, the effective area of the transmission mode inside the micro-ring resonator waveguide is 0.439 μm^2 as obtained by simulation with the R-soft software. The input peak intensity is $I_{\rm in} = P/A_{\rm eff} = 1.53\times10^8\ \mathrm{W/cm^2}$ and the amplified peak intensity is $I_A = MxI_{\rm in}$ at the resonant wavelength of micro-ring resonator. The magnification factor (M) is derived as [34],

$$M = \left|\frac{B_0}{A_0}\right|^2 = \left|\frac{-j\sin(\kappa l)\exp(-\frac{\alpha_{\rm ring}}{2}L)}{1-\cos(\kappa l)\exp(-\frac{\alpha_{\rm ring}}{2}L)}\right|^2 = \left|\frac{-x\times\sqrt{1-y^2}}{1-x\times y}\right|^2. \tag{6}$$

By substituting the parameters listed in Table I into Eq. (6), the magnification factor is estimated as 0.645 at pump wavelength, and the nonlinear refractive index of the Si-rich SiC_x is calculated as $n_2 = \Delta n/M\times I_{\rm in} = 1\times10^{-12}\ \mathrm{cm^2/W}$. Indeed, the magnification factor should be larger than 1 if the ring resonator is perfect. However, the magnification factor of resonator can be degraded due to the coupling and radiation losses of the ring-resonator beneath the bus waveguide with roughened sidewall. Unfortunately, the coupling loss is inevitable when seeding the pumping data stream from bus waveguide to ring resonator, the magnification factor is reduced when increasing the coupling factor or when roughening the sidewall during ion-beam etching process. As the ring-resonator is a key component to cause the transmission notch on the throughput spectrum, our results explain why the Kerr induced switching is less dominated than the FCA induced switching in this case. The TPA effect is dominated by the TPA induced FCA effect occurred in the bus waveguide. In previous works, the enhanced

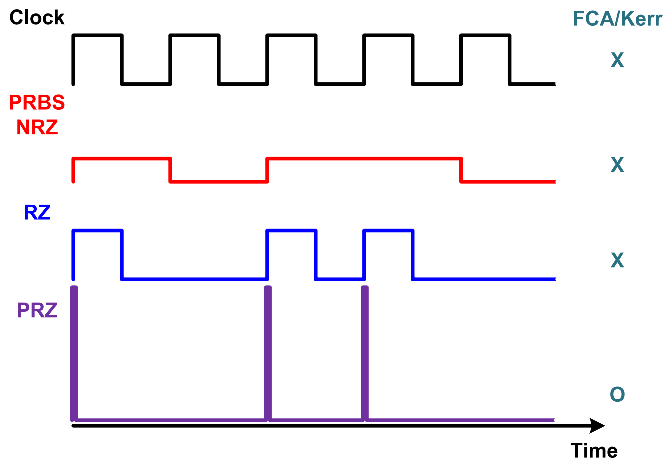

Fig. 13. The PRBS, RZ, and PRZ data format as a function of the time.

third-order nonlinear susceptibility $(\chi^{(3)})$ of one-dimensional Si nanowires has been theoretically estimated [43], indicating that the $\chi^{(3)}$ can be significantly enhanced by the quantum-confined excitons occurring in the Si nanowire. The correlation between the $\chi^{(3)}$ and effective Bohr radius (a_0) can be simply described as $\chi^{(3)} \propto (a_0)^{-6}$. For a highly confined exciton in a nanostructure, its effective Bohr radius is significantly reduced due to the increased effective hole and electron masses in the Si nanostructure. Such a nonlinear refractive index is small due to the weak nonlinear Kerr effect of the Si-rich SiC_x:Si-QD material. In this case, the all-optical switching function of the Si-rich SiCx:Si-QD bus/ring waveguide is mainly dominated by the TPA/FCA induced XAM mechanism.

D. 1.2 Gb/s All-Optical PRZ-OOK Data Inversion in SiC_x:Si-QD Based All-Optical XAM Switch

In general, the pseudorandom binary sequence (PRBS) non-return-to-zero (NRZ) data formation exhibits a bit width as wide as the period of clock signal in time domain. Even with a PRBS RZ data stream, the temporal width of one RZ bit width is only 50% of that of one NRZ bit, which cannot provide a sufficiently large peak intensity to be employed as the pumping source, as shown in Fig. 13. Hence the conventional PRBS NRZ/RZ data-stream fails to provide sufficiently high peak intensity for inducing the TPA induced FCA effect. Therefore, the pulsed RZ (PRZ) data-stream is employed in subsequent experiment, which is generated by using an arbitrary waveform generator.

To realize the limitation on modulation bandwidth of the SiC_x:Si-QD based all-optical XAM switch, the optical PRZ-OOK data stream at bit rates of 1.2 and 2.4 Gb/s are launched into the SiC_x:Si-QD based all-optical XAM switch as a pump trace for enabling wavelength conversion and data format inversion of probe beam, as shown in Fig. 14. The PRZ-OOK data bit embedded in the pump stream exhibits a pulsewidth of 83 ps, corresponding to the duty-cycles of 10% and 20% at data rates of 1.2 and 2.4 Gb/s, respectively. In particular, for investigating the response of the SiC_x:Si-QD based all-optical XAM switch to discrete and successive data bits, two specific data-streams of "010100" and "010110" are employed at both bit rates and the traces and histograms of receiving data streams are employed accordingly. By injecting the CW probe signal into the SiC_x:Si-QD based all-optical XAM switch, the probe signal can be inversely modulated by the TPA induced FCA effect to cause the XAM, which reveals a format-inverted asymmetric pulsed RZ data stream with fast rising time but slow falling time.

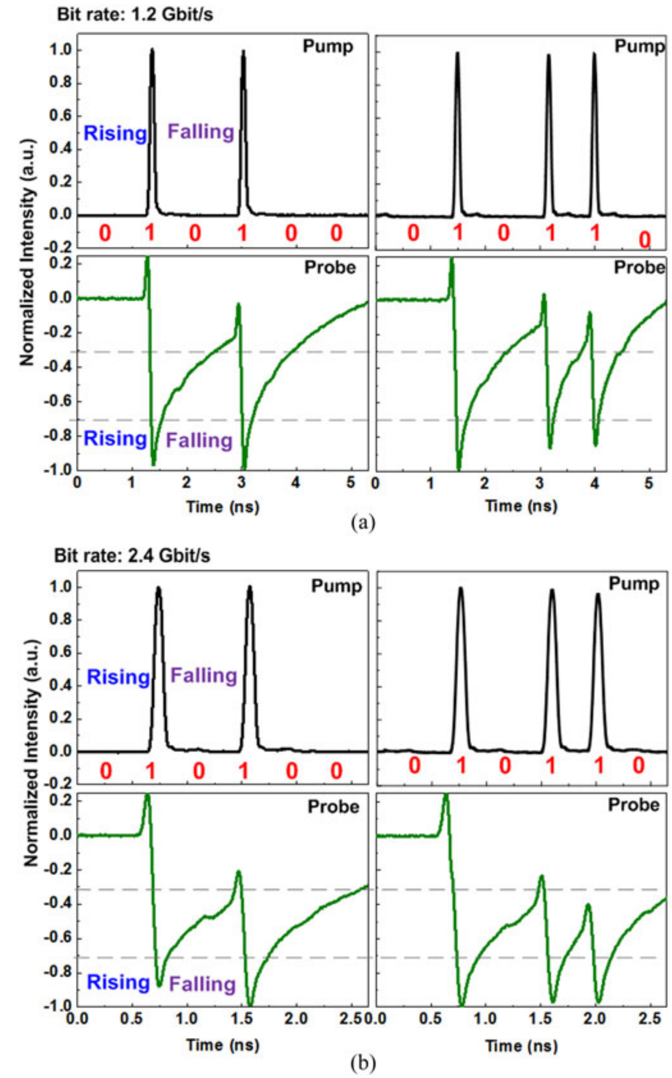

Fig. 14. The delivered PRZ-OOK data with (a) 1.2 and (b) 2.4 Gb/s in SiC_x:Si-QD based XAM switch.

The long trailing edge of the XAM response (defined as the time required for the response falling from 90 to 10% of its initial value) lengthens to 1.6 ns, indicating that the XAM in the Si-rich SiC_x:Si-QD is still limited TPA induced FCA process with finite carrier lifetime in the Si-QDs. However, both the discrete and successive on-level bits in the incoming PRZ-OOK data stream at bit rate of 1.2 Gb/s has been successfully received at the output port of the Si-rich SiCx:Si-QD based all-optical XAM switch, as shown in Fig. 14(a). For the case of discrete on-level bits of the PRZ-OOK data stream, the SiC_x:Si-QD based all-optical XAM switch perfectly converts its wavelength and inverts its format; however, the FCA induced long falling time inevitably induces a residual pedestal at the falling edge to degrade the turn-off performance. In this case, when the successive on-level bits are injected, the SiC_x:Si-QD based all-optical XAM switch inevitably shrinks its responded amplitude to the following PRZ-OOK data. This is attributed to the insufficient carriers at ground state as the relaxation of the FCA requires after first bit long response time, to diminish the pedestal at the falling edge. As a result, the SiC_x:Si-QD with insufficiently shortened carrier recovery time can partially follow up the successively incoming on-level bits at a cost of lengthened tail and decreased on/off extinction ratio. In general, the criterion of the complementary metal-oxide-semiconductor (CMOS) logic is employed judge if the data format can be decoded or not, which defines the distinguishable high and low logical levels at above 70% and below 30% of the data amplitude [44], respectively. In our case, the SiC_x:Si-QD based all-optical XAM switch is already qualified to wavelength-convert and format-invert the PRZ-OOK data at up to 1.2 Gb/s with on/off extinction ratio as high as 5.69 dB, which satisfies the CMOS criterion. When increasing the bit rate of PRZ-OOK data from 1.2 to 2.4 Gb/s, the modulated probe output from the SiC_x:Si-QD based all-optical XAM switch fails to respond the successively incoming on-level bits with qualified on and off logic levels, as shown in Fig. 14. The subsequently incoming on-bits are inevitably offset by the falling-edge pedestal of previous on-bit, which leads to a reshaped data-stream with a degraded on/off extinction ratio that can hardly reaches the criterion of CMOS required logical levels.

In more detail, the stacked histogram of the inverted PRZ-OOK data stream delivered from the SiC_x:Si-QD based XAM switch at 1.2 and 2.4 Gb/s are compared in Fig. 15. The SNR can be obtained from the stacked histogram by using the following function:

$$SNR = \frac{Ave_{1\mathrm{level}} - Ave_{0\mathrm{level}}}{\sigma_{1\mathrm{level}} + \sigma_{0\mathrm{level}}} \tag{7}$$

where the $Ave_{1\mathrm{level}}$ and $Ave_{0\mathrm{level}}$ denotes the histogram means for the 1 and 0 levels, respectively. The $\sigma_{1\mathrm{level}}$ and $\sigma_{0\mathrm{level}}$ are the standard deviations from the histogram means for the 1 and 0 levels, respectively. At 1.2 Gb/s, the inverted PRZ-OOK data exhibits a relatively clear histogram with a peak-to-peak intensity variation of 39.5%, a signal-to-noise ratio (SNR) of 4.98 dB, a rising-edge timing-jitter of 13 ps and a falling-edge lifetime of 160 ps. Obviously, the relatively low SNR results from the long falling pedestal. When increasing the bit rate of pumping PRZ-OOK data up to 2.4 Gb/s, the inverted probe PRZ-OOK data shows blurred and eye-closed histogram with undistinguishable SNR.

To compare the performance of this device with other in previous works, key parameters of similar modulators are listed in Table III. Creazzo *et al*. investigated the FCA phenomenon of SiO_x:Si-QD based slot waveguide, indicating that the carrier lifetime of SiO_x:Si-QD was as long as 30 μs characterized by time-resolved photoluminescence [26]. Navarro-Urrios *et al*. also demonstrated FCA mechanisms in the SiO_x:Si-QD based rib waveguide with shortened carrier lifetime of 20 μs [24]. Recently, Wu *et al*. greatly shortened the carrier lifetime of SiO_x:Si-QD to 9 μs in the SiO_x:Si-QD waveguide, thus providing the all-optical RZ-OOK data format inversion with a data rate of 200 kb/s [45]. Later on, the carrier lifetime of SiO_x:Si-QD waveguides is further shortened to 0.48 μs by reducing the Si-QD size to 1.7 nm, which significantly improves the bit rate of SiO_x:Si-QD waveguide modulator to 2 Mb/s [27]. Li *et al*. also demonstrated that the carrier lifetime of SiN_x:Si-QD can be as short as 2 ns [46], which is beneficial from enhancing the modulation bandwidth to 10 s–100 s MHz. Nevertheless, the isolated electrical property of oxide or nitride based dielectric host matrix still limits its applications on the ultrafast optical switch. To achieve high-speed modulation, the semi-conducting host matrix should be a potential candidate for enhancing the carrier diffusing capability. In our case, the semiconducting SiC has shown to be a promising alternative host matrix for shortening the carrier lifetime to 1.3 ns, which leads to the realization of both the ultrafast all-optical wavelength conversion and PRZ-OOK

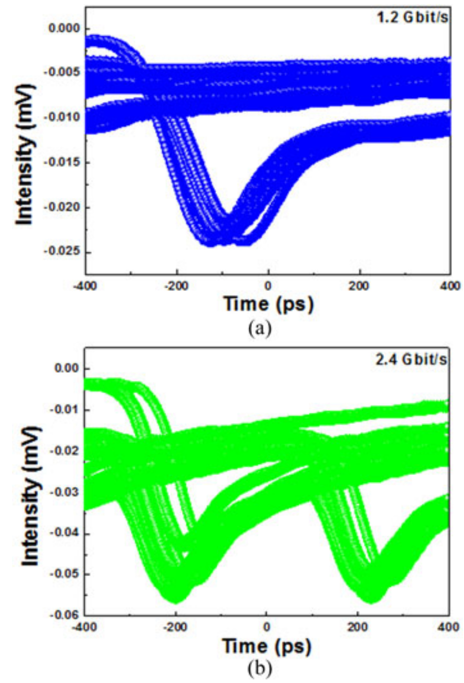

Fig. 15. (a) 1.2 and (b) 2.4 Gb/s PRZ-OOK data obtained at the probe output from the SiC_x:Si-QD based XAM switch.

TABLE III
PERFORMANCE COMPARISON OF ALL-OPTICAL MODULATION IN SI-QD BASED WAVEGUIDE

Material	Optical modulation type	Carrier lifetime	Data rate
SiO_x:Si-QD [26]	Mechanical chopper	30 μs	100 Hz
SiO_x:Si-QD [24]	Mechanical chopper	20 μs	~700 Hz
SiO_x:Si-QD [45]	Direct optical modulation	9 μs	200 kb/s
SiO_x:Si-QD [27]	Direct optical modulation	0.48 μs	2 Mb/s
SiN:Si-QD [46]	–	2 ns	–
SiC_x:Si-QD	External optical modulation	1.3 ns	1.2 Gb/s

data inversion at a bit rate of up to 1.2 Gb/s in the SiC_x:Si-QD based all-optical XAM switch. Such a new class of XAM switch has been observed and confirmed as a new kind of wavelength conversion process to enable the ultrafast optical switching.

IV. Conclusion

With the strong TPA induced FCA process, the SiC_x:Si-QD based ultrafast all-optical XAM switch which enables the format inversion of a PRZ-OOK data-stream have been demonstrated. By increasing the gap spacing between bus and micro-ring waveguides from 500 to 700 nm, the full-FWHM of the resonance dip on the SiC_x:Si-QD micro-ring resonator decreases from 116.4 to 90.2 pm, providing a Q-factor of the micro-ring resonator can be enlarged from 1.34×10^4 to 1.72×10^4. The enlarged Q-factor and the reduced coupling efficiency with optimized gap spacing of 700 nm can be compromised each other. The TPA effect occurs inside the Si-QD to generate free-carriers, which continuously absorb the probing photons at another wavelength to cause the XAM effect. The nonlinear transfer function of output power versus input power to verify the existence of TPA effect when enlarging the pumping intensity beyond 0.65 GW/cm^2. Either free carrier or the tremendous pump can induce nonlinear refractive index change to cause another Kerr switching response; however, the free-carrier induced nonlinear refractive index is trivial in the XAM case. Both the Kerr and XAM based all-optical switching responses present in the switching response concurrently, but the XAM switching still dominates the switching mechanism inside the SiC_x:Si-QD based all-optical XAM switch. With deconvolution on the power of probe at given time modulated by FCA, the carrier lifetime and dispersion factor of SiC_x:Si-QD are estimated as $\tau_c = 1.3$ ns and $\beta = 0.95$, respectively, indicating that the uniform size distribution of small-size Si-QDs is mandatory to shorten the trailing edge of the XAM and to accelerate the switching response. The nonlinear Kerr effect is existed to cause an intensity and sign reversed variable *sech* pulse response in the SiC_x:Si-QD based all-optical XAM switch. To resolve the direct all-optical modulation by nonlinear Kerr effect, its maximum is obtained by allocating the wavelength of probe beam at the original resonant dip of the SiC_x:Si-QD micro-ring, the intense pump pulse redshifts the transmission spectrum to increase the probe beam intensity. In contrast, when slightly deviating the wavelength from the resonant dip, the pump pulse induced red-shifted transmission inversely decreases the probe beam intensity and even reverse its sign. In more detail, the spacing between wavelength at positive maximal power ratio of 0.11 (at 1564.84 nm) and that at negative maximal power ratio of -0.12 (at 1564.90 nm) is about $\sim$0.06 nm, indicating that the resonant transmission dip of the micro-ring resonator must be red-shifted by 0.06 nm. Therefore, the nonlinear Kerr effect is observed in the SiC_x:Si-QD micro-ring resonator with an estimated n_2 value of 1×10^{-12} cm^2/W obtained under a peak intensity of 1.53×10^8 W/cm^2. However, the contribution of nonlinear Kerr effect is weaker than that of TPA/FCA induced XAM effect, which makes the XAM effect the dominant factor in the SiC_x:Si-QD. Therefore, the output sign of the modulated probe will be preserved or converted by the XAM effect. To realize the limitation on modulation bandwidth of the SiC_x:Si-QD based all-optical XAM switch, the optical PRZ-OOK data stream at bit rates of 1.2 or 2.4 Gb/s are launched into the SiC_x:Si-QD based all-optical XAM switch as a pump trace to enable wavelength conversion and data format inversion of probe beam. As the trailing edge of the XAM response lengthens to 1.6 ns, the Si-rich SiC_x:Si-QD is still limited by the TPA/FCA induced XAM process with finite carrier lifetime in the Si-QDs. At 1.2 Gb/s, the inverted PRZ-OOK data exhibits a relatively clear histogram with a peak-to-peak intensity variation of 39.5%, a signal-to-noise ratio (SNR) SNR of 4.98 dB, a rising-edge timing-jitter of 13 ps and a falling-edge lifetime of 160 ps. The discrete and successive on-level bits in the incoming PRZ-OOK data stream at bit rate of 1.2 Gb/s has been successfully received at the output port of the Si-rich SiC_x:Si-QD based all-optical XAM switch. When increasing the bit rate of pumping PRZ-OOK data up to 2.4 Gb/s, the inverted probe PRZ-OOK data shows blurred and eye-closed histogram with undistinguishable SNR. In summary, such a new class of XAM switching behavior has been observed in the Si-rich SiC_x:Si-QD waveguide, which is confirmed as a new kind of wavelength conversion and data-format inversion process to enable ultrafast optical switching.

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